

CORE JAVA

1. Hello World Program

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Output

Hello, World!

2. Simple Calculator

```
import java.util.Scanner;  
  
public class Calculator {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter first number: ");  
        double a = sc.nextDouble();  
  
        System.out.print("Enter second number: ");  
        double b = sc.nextDouble();  
  
        System.out.println("1.Add 2.Subtract 3.Multiply 4.Divide");  
        int choice = sc.nextInt();  
  
        switch(choice) {  
            case 1:  
                System.out.println("Result = " + (a + b));  
                break;  
            case 2:  
                System.out.println("Result = " + (a - b));  
                break;  
            case 3:  
                System.out.println("Result = " + (a * b));  
                break;  
            case 4:  
                System.out.println("Result = " + (a / b));  
                break;  
            default:  
                System.out.println("Invalid Choice");  
        }  
    }  
}
```

Output

Enter first number: 10
Enter second number: 5
1.Add 2.Subtract 3.Multiply 4.Divide
1
Result = 15.0

3. Even or Odd Checker

```
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if(n % 2 == 0)
            System.out.println("Even");
        else
            System.out.println("Odd");
    }
}
```

Output

Enter a number: 8
Even

4. Leap Year Checker

```
import java.util.Scanner;

public class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter year: ");
        int year = sc.nextInt();

        if((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
            System.out.println("Leap Year");
        else
            System.out.println("Not a Leap Year");
    }
}
```

Output

Enter year: 2024
Leap Year

5. Multiplication Table

```
import java.util.Scanner;

public class MultiplicationTable {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        int n = sc.nextInt();

        for(int i=1;i<=10;i++) {
            System.out.println(n + " x " + i + " = " + (n*i));
        }
    }
}
```

Output

Enter number: 5
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50

6. Data Type Demonstration

```
public class DataTypes {
    public static void main(String[] args) {

        int num = 100;
        float f = 12.5f;
        double d = 123.456;
        char ch = 'A';
        boolean flag = true;

        System.out.println("Int = " + num);
        System.out.println("Float = " + f);
        System.out.println("Double = " + d);
        System.out.println("Char = " + ch);
        System.out.println("Boolean = " + flag);
    }
}
```

Output

Int = 100
Float = 12.5
Double = 123.456
Char = A
Boolean = true

7. Type Casting Example

```
public class TypeCasting {  
    public static void main(String[] args) {  
  
        double d = 45.89;  
        int i = (int)d;  
  
        int num = 20;  
        double d2 = (double)num;  
  
        System.out.println("Double = " + d);  
        System.out.println("Int = " + i);  
  
        System.out.println("Int = " + num);  
        System.out.println("Double = " + d2);  
    }  
}
```

Output

Double = 45.89
Int = 45
Int = 20
Double = 20.0

8. Operator Precedence

```
public class OperatorPrecedence {  
    public static void main(String[] args) {  
  
        int result1 = 10 + 5 * 2;  
        int result2 = (10 + 5) * 2;  
  
        System.out.println(result1);  
        System.out.println(result2);  
    }  
}
```

Output

20
30

9. Grade Calculator

Code

```
import java.util.Scanner;

public class GradeCalculator {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks: ");
        int marks = sc.nextInt();

        if(marks >= 90)
            System.out.println("Grade A");
        else if(marks >= 80)
            System.out.println("Grade B");
        else if(marks >= 70)
            System.out.println("Grade C");
        else if(marks >= 60)
            System.out.println("Grade D");
        else
            System.out.println("Grade F");
    }
}
```

Output

Enter marks: 85
Grade B

10. Number Guessing Game

```
import java.util.Random;
import java.util.Scanner;

public class GuessNumber {
    public static void main(String[] args) {

        Random rand = new Random();
        int number = rand.nextInt(100) + 1;

        Scanner sc = new Scanner(System.in);
        int guess;
```

```

do {
    System.out.print("Guess a number (1-100): ");
    guess = sc.nextInt();

    if(guess > number)
        System.out.println("Too High");
    else if(guess < number)
        System.out.println("Too Low");
    else
        System.out.println("Correct!");
} while(guess != number);
}
}

```

Output

Guess a number (1-100): 50
Too High

Guess a number (1-100): 25
Too Low

Guess a number (1-100): 35
Correct!

11. Factorial Calculator

```

import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        long fact = 1;

        for(int i = 1; i <= n; i++) {
            fact = fact * i;
        }

        System.out.println("Factorial = " + fact);
    }
}

```

Output

Enter a number: 5
Factorial = 120

12. Method Overloading

```
public class MethodOverloading {  
  
    static int add(int a, int b) {  
        return a + b;  
    }  
  
    static double add(double a, double b) {  
        return a + b;  
    }  
  
    static int add(int a, int b, int c) {  
        return a + b + c;  
    }  
  
    public static void main(String[] args) {  
  
        System.out.println(add(10, 20));  
        System.out.println(add(5.5, 4.5));  
        System.out.println(add(10, 20, 30));  
    }  
}
```

Output

30
10.0
60

13. Recursive Fibonacci

```
import java.util.Scanner;  
  
public class Fibonacci {  
  
    static int fibonacci(int n) {  
        if(n <= 1)  
            return n;  
  
        return fibonacci(n - 1) + fibonacci(n - 2);  
    }  
  
    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

System.out.print("Enter n: ");
int n = sc.nextInt();

System.out.println("Fibonacci Number = " + fibonacci(n));
}
}

```

Output

```

Enter n: 7
Fibonacci Number = 13

```

14. Array Sum and Average

```

import java.util.Scanner;

public class ArraySumAverage {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int arr[] = new int[n];

        int sum = 0;

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        double average = (double) sum / n;

        System.out.println("Sum = " + sum);
        System.out.println("Average = " + average);
    }
}

```

Output

```

Enter number of elements: 5
10 20 30 40 50
Sum = 150
Average = 30.0

```


15. String Reversal

```
import java.util.Scanner;

public class StringReverse {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String reversed =
            new StringBuilder(str).reverse().toString();

        System.out.println("Reversed String = " + reversed);
    }
}
```

Output

```
Enter a string: Java
Reversed String = avaJ
```

16. Palindrome Checker

```
import java.util.Scanner;

public class Palindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        str = str.replaceAll("[^a-zA-Z0-9]", "")
            .toLowerCase();

        String reverse =
            new StringBuilder(str).reverse().toString();

        if(str.equals(reverse))
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```

Output

Enter a string: madam
Palindrome

17. Class and Object Creation

```
class Car {  
  
    String make;  
    String model;  
    int year;  
  
    void displayDetails() {  
        System.out.println("Make : " + make);  
        System.out.println("Model : " + model);  
        System.out.println("Year : " + year);  
    }  
}  
  
public class CarDemo {  
  
    public static void main(String[] args) {  
  
        Car c1 = new Car();  
  
        c1.make = "Toyota";  
        c1.model = "Corolla";  
        c1.year = 2024;  
  
        c1.displayDetails();  
    }  
}
```

Output

Make : Toyota
Model : Corolla
Year : 2024

18. Inheritance Example

```
class Animal {  
  
    void makeSound() {  
        System.out.println("Animal Sound");  
    }  
}  
  
class Dog extends Animal {
```

```

    @Override
    void makeSound() {
        System.out.println("Bark");
    }
}

public class InheritanceDemo {

    public static void main(String[] args) {

        Animal a = new Animal();
        Dog d = new Dog();

        a.makeSound();
        d.makeSound();
    }
}

```

Output

Animal Sound
Bark

19. Interface Implementation

```

interface Playable {
    void play();
}

class Guitar implements Playable {

    public void play() {
        System.out.println("Playing Guitar");
    }
}

class Piano implements Playable {

    public void play() {
        System.out.println("Playing Piano");
    }
}

public class InterfaceDemo {

    public static void main(String[] args) {

        Guitar g = new Guitar();
        Piano p = new Piano();
    }
}

```

```
        g.play();
        p.play();
    }
}
```

Output

Playing Guitar
Playing Piano

20. Try-Catch Example

```
import java.util.Scanner;

public class TryCatchDemo {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        try {
            int result = a / b;
            System.out.println("Result = " + result);
        }
        catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero");
        }
    }
}
```

Output

Enter first number: 10
Enter second number: 0
Cannot divide by zero

21. Custom Exception

```
import java.util.Scanner;

class InvalidAgeException extends Exception {
    InvalidAgeException(String message) {
        super(message);
    }
}
```

```

    }
}

public class CustomExceptionDemo {

    static void checkAge(int age) throws InvalidAgeException {
        if(age < 18)
            throw new InvalidAgeException("Age must be 18 or above.");
        else
            System.out.println("Eligible to Vote");
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Age: ");
        int age = sc.nextInt();

        try {
            checkAge(age);
        }
        catch(InvalidAgeException e) {
            System.out.println(e.getMessage());
        }
    }
}

```

Output

```

Enter Age: 16
Age must be 18 or above.

```

22. File Writing

```

import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class FileWriting {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Text: ");
        String text = sc.nextLine();

        try {
            FileWriter fw = new FileWriter("output.txt");

```

```

        fw.write(text);
        fw.close();

        System.out.println("Data Written Successfully");
    }
    catch(IOException e) {
        System.out.println(e);
    }
}
}

```

Output

Enter Text: Hello Java
Data Written Successfully

23. File Reading

```

import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class FileReading {

    public static void main(String[] args) {

        try {
            BufferedReader br =
                new BufferedReader(
                    new FileReader("output.txt"));

            String line;

            while((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();
        }
        catch(IOException e) {
            System.out.println(e);
        }
    }
}

```

Output

Hello Java

24. ArrayList Example

```
import java.util.ArrayList;
import java.util.Scanner;

public class ArrayListDemo {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ArrayList<String> students =
            new ArrayList<>();

        System.out.print("How many names? ");
        int n = sc.nextInt();
        sc.nextLine();

        for(int i = 0; i < n; i++) {
            System.out.print("Enter Name: ");
            students.add(sc.nextLine());
        }

        System.out.println("\nStudent Names");

        for(String name : students) {
            System.out.println(name);
        }
    }
}
```

Output

```
How many names? 3
Enter Name: Ram
Enter Name: Sam
Enter Name: Tom
```

```
Student Names
Ram
Sam
Tom
```

25. HashMap Example

```
import java.util.HashMap;
import java.util.Scanner;

public class HashMapDemo {
```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    HashMap<Integer,String> students =
        new HashMap<>();

    System.out.print("Number of Students: ");
    int n = sc.nextInt();

    for(int i = 0; i < n; i++) {

        System.out.print("Enter ID: ");
        int id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Name: ");
        String name = sc.nextLine();

        students.put(id, name);
    }

    System.out.print("Search ID: ");
    int searchId = sc.nextInt();

    System.out.println("Student Name: "
        + students.get(searchId));
    }
}

```

Output

Number of Students: 2

Enter ID: 101

Enter Name: Ram

Enter ID: 102

Enter Name: Sam

Search ID: 102

Student Name: Sam

26. Thread Creation

```

class MyThread extends Thread {

    public void run() {

```



```

        for(int i = 1; i <= 5; i++) {
            System.out.println(
                Thread.currentThread().getName()
                + " Running");
        }
    }
}

public class ThreadDemo {

    public static void main(String[] args) {

        MyThread t1 = new MyThread();
        MyThread t2 = new MyThread();

        t1.start();
        t2.start();
    }
}

```

Output

```

Thread-0 Running
Thread-1 Running
Thread-0 Running
Thread-1 Running

```

27. Lambda Expressions

```

import java.util.*;

public class LambdaSort {

    public static void main(String[] args) {

        List<String> list =
            Arrays.asList("Java","Python","C","Ruby");

        Collections.sort(list,
            (a,b) -> a.compareTo(b));

        System.out.println(list);
    }
}

```

Output

```

[C, Java, Python, Ruby]

```

28. Stream API

```
import java.util.*;
import java.util.stream.*;

public class StreamDemo {

    public static void main(String[] args) {

        List<Integer> numbers =
            Arrays.asList(1,2,3,4,5,6,7,8,9,10);

        List<Integer> even =
            numbers.stream()
                .filter(n -> n % 2 == 0)
                .collect(Collectors.toList());

        System.out.println(even);
    }
}
```

Output

[2, 4, 6, 8, 10]

29. Records (Java 16+)

```
import java.util.List;

record Person(String name, int age) {}

public class RecordDemo {

    public static void main(String[] args) {

        Person p1 = new Person("Ram", 20);
        Person p2 = new Person("Sam", 16);

        List<Person> persons =
            List.of(p1, p2);

        persons.stream()
            .filter(p -> p.age() >= 18)
            .forEach(System.out::println);
    }
}
```

Output

Person[name=Ram, age=20]

30. Pattern Matching for Switch (Java 21)

```
public class PatternSwitch {  
  
    static void checkType(Object obj) {  
  
        switch(obj) {  
  
            case Integer i ->  
                System.out.println(  
                    "Integer: " + i);  
  
            case String s ->  
                System.out.println(  
                    "String: " + s);  
  
            case Double d ->  
                System.out.println(  
                    "Double: " + d);  
  
            default ->  
                System.out.println(  
                    "Unknown Type");  
        }  
    }  
  
    public static void main(String[] args) {  
  
        checkType(100);  
        checkType("Java");  
        checkType(25.5);  
    }  
}
```

Output

```
Integer: 100  
String: Java  
Double: 25.5
```

31. Basic JDBC Connection

Code

```
import java.sql.*;  
  
public class JDBCSelect {  
  
    public static void main(String[] args) {
```

```

String url =
    "jdbc:mysql://localhost:3306/college";
String user = "root";
String password = "root";

try {
    Connection con =
        DriverManager.getConnection(
            url, user, password);

    Statement st =
        con.createStatement();

    ResultSet rs =
        st.executeQuery(
            "SELECT * FROM students");

    while(rs.next()) {
        System.out.println(
            rs.getInt("id")
            + " "
            + rs.getString("name"));
    }

    con.close();
}
catch(Exception e) {
    System.out.println(e);
}
}
}

```

Output

```

101 Ram
102 Sam
103 Tom

```

32. Insert and Update Operations in JDBC

```

import java.sql.*;

public class StudentDAO {

    static String url =
        "jdbc:mysql://localhost:3306/college";

    static String user = "root";
    static String password = "root";

```

```

public static void insertStudent(
    int id, String name)
    throws Exception {

    Connection con =
        DriverManager.getConnection(
            url,user,password);

    PreparedStatement ps =
        con.prepareStatement(
            "INSERT INTO students VALUES(?,?)");

    ps.setInt(1,id);
    ps.setString(2,name);

    ps.executeUpdate();

    System.out.println(
        "Student Inserted");

    con.close();
}

public static void updateStudent(
    int id, String name)
    throws Exception {

    Connection con =
        DriverManager.getConnection(
            url,user,password);

    PreparedStatement ps =
        con.prepareStatement(
            "UPDATE students SET name=? WHERE id=?");

    ps.setString(1,name);
    ps.setInt(2,id);

    ps.executeUpdate();

    System.out.println(
        "Student Updated");

    con.close();
}

public static void main(String[] args)
    throws Exception {

```

```

        insertStudent(104,"John");
        updateStudent(104,"David");
    }
}

```

Output

Student Inserted
Student Updated

33. Transaction Handling in JDBC

```

import java.sql.*;

public class MoneyTransfer {

    public static void main(String[] args) {

        try {

            Connection con =
                DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/bank",
                    "root",
                    "root");

            con.setAutoCommit(false);

            PreparedStatement debit =
                con.prepareStatement(
                    "UPDATE accounts SET balance=balance-? WHERE id=?");

            PreparedStatement credit =
                con.prepareStatement(
                    "UPDATE accounts SET balance=balance+? WHERE id=?");

            debit.setDouble(1,500);
            debit.setInt(2,1);

            credit.setDouble(1,500);
            credit.setInt(2,2);

            debit.executeUpdate();
            credit.executeUpdate();

            con.commit();

            System.out.println(
                "Transfer Successful");
        }
    }
}

```

```
        catch(Exception e) {
            System.out.println(
                "Transaction Failed");
        }
    }
}
```

Output

Transfer Successful

34. Create and Use Java Modules

module-info.java (com.utils)

```
module com.utils {
    exports com.utils;
}
```

Utility Class

```
package com.utils;

public class Message {

    public static void show() {
        System.out.println(
            "Welcome to Modules");
    }
}
```

module-info.java (com.greetings)

```
module com.greetings {
    requires com.utils;
}
```

Main Class

```
package com.greetings;

import com.utils.Message;

public class Main {

    public static void main(String[] args) {
        Message.show();
    }
}
```

Output

Welcome to Modules

35. TCP Client-Server Chat

Server Program

```
import java.net.*;
import java.io.*;

public class Server {

    public static void main(String[] args)
        throws Exception {

        ServerSocket ss =
            new ServerSocket(5000);

        System.out.println(
            "Waiting for Client...");

        Socket s = ss.accept();

        BufferedReader br =
            new BufferedReader(
                new InputStreamReader(
                    s.getInputStream()));

        String msg = br.readLine();

        System.out.println(
            "Client : " + msg);

        ss.close();
    }
}
```

Client Program

```
import java.net.*;
import java.io.*;

public class Client {

    public static void main(String[] args)
        throws Exception {

        Socket s =
            new Socket("localhost",5000);
```



```

        PrintWriter pw =
            new PrintWriter(
                s.getOutputStream(),
                true);

        pw.println("Hello Server");

        s.close();
    }
}

```

Output

Waiting for Client...

Client : Hello Server

36. HTTP Client API (Java 11+)

```

import java.net.URI;
import java.net.http.*;

public class HttpDemo {

    public static void main(String[] args)
        throws Exception {

        HttpClient client =
            HttpClient.newHttpClient();

        HttpRequest request =
            HttpRequest.newBuilder()
                .uri(
                    URI.create(
                        "https://api.github.com"))
                .build();

        HttpResponse<String> response =
            client.send(
                request,
                HttpResponse.BodyHandlers
                    .ofString());

        System.out.println(
            response.statusCode());

        System.out.println(
            response.body());
    }
}

```

Output

```
200
{ JSON Response }
```

37. Using javap to Inspect Bytecode

Program

```
public class Test {

    public void display() {
        System.out.println("Hello");
    }
}
```

Compile

```
javac Test.java
```

Inspect Bytecode

```
javap -c Test
```

Output

```
public void display();
Code:
0: getstatic
3: ldc
5: invokevirtual
8: return
```

38. Decompile a Class File

Program

```
public class Demo {

    public static void main(String[] args) {
        System.out.println(
            "Java Decompiler");
    }
}
```

Steps

1. Compile:
javac Demo.java

2. Open Demo.class
using JD-GUI or CFR
3. Decompiled code
will resemble original source.

Output

```
public class Demo {  
    public static void main(  
        String[] args) {  
  
        System.out.println(  
            "Java Decompiler");  
    }  
}
```

39. Reflection in Java

```
import java.lang.reflect.*;  
  
class Sample {  
  
    public void show() {  
        System.out.println(  
            "Reflection Demo");  
    }  
}  
  
public class ReflectionDemo {  
  
    public static void main(String[] args)  
        throws Exception {  
  
        Class<?> c =  
            Class.forName("Sample");  
  
        Method methods[] =  
            c.getDeclaredMethods();  
  
        for(Method m : methods) {  
            System.out.println(  
                m.getName());  
        }  
  
        Object obj =  
            c.getDeclaredConstructor()  
                .newInstance();  
  
        Method m =  
            c.getMethod("show");
```

```
        m.invoke(obj);
    }}
```

Output

show
Reflection Demo

40. Virtual Threads (Java 21)

Code

```
public class VirtualThreadDemo {

    public static void main(String[] args)
        throws Exception {

        for(int i=1;i<=100000;i++) {

            Thread.startVirtualThread(
                () -> System.out.println(
                    "Virtual Thread"));
        }

        Thread.sleep(2000);
    }
}
```

Output

Virtual Thread
Virtual Thread
Virtual Thread

41. Executor Service and Callable

```
import java.util.concurrent.*;

public class ExecutorDemo {

    public static void main(String[] args)
        throws Exception {

        ExecutorService service =
            Executors.newFixedThreadPool(3);

        Callable<Integer> task1 =
            () -> 10;

        Callable<Integer> task2 =
```

```
    () -> 20;

    Future<Integer> f1 =
        service.submit(task1);

    Future<Integer> f2 =
        service.submit(task2);

    System.out.println(
        "Result 1 = "
        + f1.get());

    System.out.println(
        "Result 2 = "
        + f2.get());

    service.shutdown();
}
}
```

Output

```
Result 1 = 10
Result 2 = 20
```